

EXPLAINABLE AI FRAMEWORK FOR CLINICALLY TRUSTWORTHY MEDICAL IMAGE CLASSIFICATION

An Internship Report

submitted in partial fulfillment of the requirements for the award of the degree of

BACHELOR OF TECHNOLOGY

IN

COMPUTER SCIENCE & ENGINEERING

Submitted by

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

**LAKIREDDY BALIREDDY COLLEGE OF ENGINEERING
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2025-2026



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DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING

CERTIFICATE

This is to certify that internship work entitled “**Explainable AI Framework for Clinically Trustworthy Medical Image Classification**” the bonafide work done by **JAJULA KISHORE NANDHAN (22761A0588)** in partial fulfillment of the requirements for the award of the degree in **BACHELOR OF TECHNOLOGY** in Computer Science and Engineering from **LAKIREDDY BALI REDDY COLLEGE OF ENGINEERING, MYLAVARAM** during the year 2025-2026.

D. Anil kumar
(Internship Coordinators)

Dr. S. Nagarjuna Reddy
(Head of the Department)

EXTERNAL EXAMINER

CERTIFICATE OF COMPLETION

ACKNOWLEDGEMENT

I express my sincere thanks to the management for their support in completing my internship work. I also express my heartfelt gratitude and deep sense of respect to our Principal, **Dr. K. Appa Rao**, for providing all the necessary assistance and for his constant encouragement and inspiration to carry out this work in the institute.

I take this opportunity to convey my sincere thanks to **Dr. S. Nagarjuna Reddy**, Head of the Department of CSE, for his continuous encouragement, cooperation, and valuable support which contributed greatly to the successful completion of my work.

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I also owe my sincere appreciation to all those who directly or indirectly helped me in the completion of this work. My special thanks to the **Librarian of LBRCE** for providing access to valuable books and reference materials.

Finally, I am deeply grateful to all the **teaching and non-teaching staff of the CSE Department** for their constant support and cooperation. I take this opportunity to express my profound gratitude to my **parents** for their continuous encouragement and blessings, which have always been my greatest source of strength.

JAJULA KISHORE NANDHAN

22761A0588

DECLARATION

We hereby declare that the internship report entitled “**Explainable AI Framework for Clinically Trustworthy Medical Image Classification**” is submitted to JNTUK in partial fulfillment of the requirements for the award of the degree of **Bachelor of Technology (B.Tech)**.

This report is an original work carried out by us, and the matter embodied in this internship report is genuine. It has not been submitted earlier to this university or any other university for the award of any degree, diploma, or prize.

SUBMITTED BY

JAJULA KISHORE NANDHAN

Executive Summary

This report presents the work carried out during my internship at **NIT Warangal**, undertaken under the expert guidance of **Mr. E. Suresh Babu**, during the period 15/10/2025 to 18/11/2025. The internship focused on the development of an **Explainable Artificial Intelligence (XAI) framework** aimed at improving clinical trust, transparency, and interpretability in AI-based medical image classification systems.

During this internship, I worked extensively on designing and integrating multiple XAI methodologies such as **Grad-CAM/Grad-CAM++**, **SHAP/TreeSHAP**, **Integrated Gradients**, **Attention Rollout**, **Prototype-based explanations**, and counterfactual reasoning. These techniques were applied across deep learning architectures including **2D slice encoders**, **3D volumetric models**, and **Multiple Instance Learning (MIL) / Transformer aggregators** to create a robust and interpretable diagnostic pipeline. The project also involved building a **clinician-oriented rendering interface** capable of producing heatmaps, saliency overlays, confidence-calibrated predictions, feature attributions, and DICOM-compatible explanation outputs.

Through this internship, I gained strong practical exposure to **medical imaging workflows**, **preprocessing and harmonization pipelines**, **model training and fusion strategies**, and advanced **explainability evaluation metrics** such as sufficiency/comprehensiveness tests, insertion-deletion curves, stability checks, uncertainty-aware saliency, and IoU-based explanation validity. Additionally, I worked on governance and documentation aspects aligned with **TRIPOD-AI/CONSORT-AI** standards to ensure transparency, reproducibility, and auditability of the AI system.

Overall, this internship significantly strengthened my technical capabilities in deep learning, model interpretability, and clinical AI development. It enhanced my problem-solving skills, research orientation, and ability to design trustworthy AI solutions for healthcare. The experience has prepared me to contribute effectively to real-world medical AI systems that require high levels of reliability, explanation quality, and clinical usability.

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1. INTRODUCTION

Objective of the Internship:

The primary objective of this internship at **NIT Warangal**, carried out under the guidance of **Mr. E. Suresh Babu**, is to bridge the gap between theoretical knowledge and real-world research practices in the field of **Explainable Artificial Intelligence (XAI)** for medical imaging. The internship focuses on developing an interpretable and clinically trustworthy AI framework that enhances transparency in deep learning–based diagnostic systems.

The project aims to integrate multiple explainability techniques—such as **Grad-CAM**, **SHAP**, **Attention Rollout**, **Integrated Gradients**, and prototype-based methods—into standard 2D, 3D, and MIL-based classifiers. Through this work, the internship emphasizes understanding the complete AI pipeline, from **data preprocessing and model development** to **explainability evaluation**, **clinical rendering**, and **governance compliance** aligned with TRIPOD-AI and CONSORT-AI guidelines.

This internship also enabled the development of strong conceptual and practical skills in **medical imaging**, **XAI methodology**, and **trust validation techniques**, contributing to the creation of AI systems that are transparent, reliable, and usable by clinicians in high-stakes environments.

Company Overview:

National Institute of Technology (NIT) Warangal is one of India’s premier engineering and research institutions, renowned for its contributions to science, technology, and innovation. The institute fosters advanced research across multiple domains, including Artificial Intelligence, Medical Imaging, Data Science, and Healthcare Technologies.

The internship was conducted as part of NIT Warangal’s research-oriented learning initiative, which provides students the opportunity to work on cutting-edge problems aligned with current industry and healthcare demands. Under the mentorship of experienced faculty, the program encourages hands-on experimentation, critical thinking, and the development of research-grade solutions.

Through this internship framework, NIT Warangal enables students to apply their academic foundations to meaningful real-world challenges—such as improving transparency and trust in AI-driven clinical diagnosis—thereby strengthening both technical competency and research orientation.

Relevance to My Field:

As a Computer Science and Engineering student, this internship aligns directly with emerging domains such as **Deep Learning**, **Medical AI**, **Explainability**, and **Model Interpretability**, which are increasingly vital in modern healthcare applications. The project provided valuable exposure to:

- Clinical workflows and radiology diagnostic systems
- AI fairness, trust, and transparency requirements
- Integration of XAI methods with modern deep learning architectures
- Practical challenges in deploying AI safely in healthcare

This experience strengthened my technical foundation and research capabilities, preparing me for future roles in **AI for healthcare**, **machine learning engineering**, and **explainable deep learning systems**.

2. INTERNSHIP ACTIVITIES

2.1 Role and Responsibilities

During my internship at **NIT Warangal**, under the guidance of **Mr. E. Suresh Babu**, I worked as an **Explainable AI Intern**. My role mainly involved understanding medical imaging workflows, implementing explainability techniques, and integrating them with deep learning models to improve clinical trust and transparency.

As part of my responsibilities, I worked on applying methods such as **Grad-CAM**, **SHAP**, **Integrated Gradients**, and **Attention-based explanations** to highlight important regions in CT scan predictions. I assisted in preparing datasets, preprocessing medical images, and evaluating model outputs using trust-based metrics. I also contributed to generating visual explanation panels and validating the reliability of these explanations through different comparison techniques.

Throughout the internship, I followed systematic research practices, documented observations, and ensured each explainability method was tested and reviewed properly. This helped me build a solid foundation in explainable AI and medical image interpretation.

2.2 Modules Undertaken

During this internship at **NIT Warangal**, I worked on a series of **twelve structured modules**, each focusing on a specific stage of building an Explainable AI (XAI) framework for medical image classification. These modules covered data preparation, model development, explainability integration, trust validation, and deployment-oriented reporting. Each module was designed to strengthen practical skills, simulate real-world clinical AI workflows, and build an end-to-end explainable diagnostic system.

Module 1: Data Governance & Curation

Activities Performed:

- Collected and organized medical imaging datasets.
- Performed de-identification checks and maintained metadata.
- Created structured dataset splits for training and evaluation.

Skills & Learning Outcomes:

- Understanding of dataset hygiene, ethics, and reproducibility.
- Ability to maintain site-aware and structured medical image datasets.

Module 2: Preprocessing & Harmonization

Activities Performed:

- Resampled CT volumes to standard spacing.
- Applied lung segmentation and region cropping.
- Implemented HU windowing for consistent image quality.

Skills & Learning Outcomes:

- Improved understanding of medical image normalization and preprocessing.

Module 3: Baseline 2D Slice Classification

Activities Performed:

- Trained a 2D CNN on CT slices.
- Used balanced losses and embedding extraction.

Skills & Learning Outcomes:

- Learned fundamentals of model training and slice-level classification.

Module 4: MIL / Transformer Study-Level Aggregation

Activities Performed:

- Applied MIL/Transformer methods to aggregate slice embeddings.
- Visualized attention maps demonstrating key slices.

Skills & Learning Outcomes:

- Understood study-level reasoning and attention mechanisms.

Module 5: Lightweight 3D Encoder

Activities Performed:

- Built a 3D CNN for volumetric CT analysis.
- Managed memory and optimized training performance.

Skills & Learning Outcomes:

- Gained exposure to 3D deep learning and volumetric modeling.

Module 6: Hybrid Fusion (2D + 3D)

Activities Performed:

- Combined 2D and 3D models using fusion strategies.
- Calibrated final predictions for reliability.

Skills & Learning Outcomes:

- Learned multi-modal model fusion and calibration techniques.

Module 7: XAI – Saliency Maps (Grad-CAM Family)

Activities Performed:

- Generated Grad-CAM/Grad-CAM++ maps for both 2D and 3D models.
- Ensured consistency using proper color mapping and overlays.

Skills & Learning Outcomes:

- Strengthened practical experience with saliency-based explainability.

Module 8: XAI – SHAP & Prototype Explanations

Activities Performed:

- Applied SHAP to quantify feature importance.
- Implemented prototype-based reasoning for intuitive comparisons.

Skills & Learning Outcomes:

- Understood feature attribution and case-based interpretability.

Module 9: Explanation Validity Metrics

Activities Performed:

- Evaluated sufficiency, comprehensiveness, insertion–deletion curves.
- Compared heatmaps with lesion masks for quality measurement.

Skills & Learning Outcomes:

- Learned how to measure explanation trustworthiness quantitatively.

Module 10: Robustness & Domain Generalization

Activities Performed:

- Applied augmentation, noise, and blur tests.
- Improved model stability across dataset variations.

Skills & Learning Outcomes:

- Understood how robustness affects clinical reliability.

Module 11: Clinician-Facing Report & DICOM Export

Activities Performed:

- Generated heatmap overlays and feature summaries.
- Created PDF reports and DICOM-Seg/SR outputs for clinical use.

Skills & Learning Outcomes:

- Learned medical reporting formats and clinician-friendly visualization.

Module 12: Reproducibility, Model Cards & Governance

Activities Performed:

- Documented model details, datasets, metrics, and configuration files.
- Prepared model cards following TRIPOD-AI/CONSORT-AI guidelines.

Skills & Learning Outcomes:

- Built a complete understanding of transparency and auditing for AI systems.

3.SKILLS AND KNOWLEDGE GAINED

During my internship at **NIT Warangal**, I gained valuable technical knowledge and practical experience in the field of **Explainable AI (XAI)** and medical image analysis. I learned how to develop interpretable deep learning models by integrating explainability techniques such as **Grad-CAM, SHAP, Integrated Gradients**, and **attention-based visualizations**, enabling transparent and clinically meaningful insights from AI predictions.

Through the different modules, I acquired hands-on experience in preprocessing medical imaging datasets, building 2D and 3D deep learning models, and designing study-level aggregation methods like **Multiple Instance Learning (MIL)** and transformer-based attention. This helped me understand how slice-level and volume-level features contribute to diagnostic decisions and how explainability can highlight these contributions effectively.

Working on explanation validity metrics strengthened my understanding of how to evaluate trustworthiness using methods like **sufficiency, comprehensiveness, insertion–deletion curves**, and stability checks. I also learned how to generate clinician-friendly outputs such as heatmap overlays, SHAP summaries, DICOM-Seg/SR files, and diagnostic reports that support radiologists in interpreting AI-generated results.

In addition to technical expertise, the internship improved my problem-solving skills, research discipline, and ability to analyze complex model behaviors. Completing multiple modules within defined milestones helped build my time management and self-learning capabilities. Overall, this internship provided me with a comprehensive understanding of explainable deep learning, medical image processing, and the professional practices required for building responsible AI systems for healthcare.

4. CHALLENGES AND HOW I OVERCAME THEM

During my internship, I encountered several challenges that helped me grow both technically and personally. One of the major challenges was understanding the complexity of medical imaging data and the different preprocessing steps required, such as lung segmentation, standardizing voxel spacing, and handling variations in CT scan quality. To overcome this, I spent additional time studying medical image processing fundamentals and practiced using multiple preprocessing tools until I became more comfortable.

Another challenge I faced was implementing and interpreting different explainability techniques. Methods like Grad-CAM, SHAP, and Integrated Gradients required a clear understanding of model internals and how each method visualizes feature importance. I resolved this by experimenting with smaller models first, comparing different outputs, and referring to research papers and documentation to understand their behavior.

Working with 3D models and attention-based MIL architectures also posed difficulties due to higher computational requirements and more complex training workflows. I overcame this by optimizing model configurations, using lighter architectures during experimentation, and seeking guidance whenever I encountered training bottlenecks.

Time management was another challenge, as the internship involved completing multiple modules within planned milestones. To handle this, I created a structured timeline, broke down tasks into smaller achievable goals, and maintained steady progress each week.

These challenges not only improved my technical understanding but also strengthened my confidence, discipline, and problem-solving approach—skills that are essential for working on real-world AI and research-oriented projects.

5. CONCLUSION

The Explainable AI internship was an enriching and meaningful learning experience that allowed me to explore the intersection of deep learning, medical imaging, and model interpretability. Throughout the internship, I gained a solid understanding of how explainability techniques such as Grad-CAM, SHAP, Integrated Gradients, and attention-based methods can be integrated into diagnostic models to improve transparency and clinical trust. Working through structured modules helped me build practical skills in preprocessing medical images, training 2D and 3D models, implementing explainability pipelines, and evaluating the quality of explanations using trust-based metrics.

This internship strengthened not only my technical abilities but also my analytical thinking, problem-solving skills, and research discipline. I learned how to interpret model behavior, identify challenges in medical AI workflows, and design outputs that can meaningfully support clinicians in decision-making. It also gave me a broader understanding of how responsible AI systems are developed, validated, and prepared for real-world use in sensitive domains like healthcare.

Overall, this internship provided valuable insights into explainable deep learning and medical imaging, helped deepen my interest in AI for healthcare, and has been an important milestone in my academic and professional journey. It has prepared me to take on future challenges in building trustworthy and reliable AI systems.

7. LEARNING OUTCOMES

The Explainable AI internship provided me with a broad range of learning experiences that contributed significantly to both my technical and professional development. Throughout this internship, I developed a strong foundation in medical image processing, deep learning–based classification, and modern explainability techniques used in clinical AI systems.

By completing modules such as saliency-based XAI (Grad-CAM, Integrated Gradients), feature attribution (SHAP), attention-based modeling, and 2D–3D hybrid architectures, I gained hands-on experience in building interpretable AI pipelines. I learned how to preprocess CT scans effectively, train slice-level and volume-level models, generate trustworthy explanations, and validate their reliability using evaluation metrics such as sufficiency, comprehensiveness, and insertion–deletion curves.

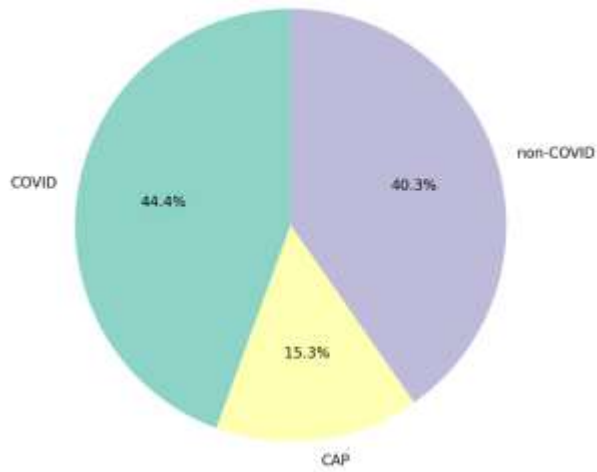
The internship also improved my analytical and problem-solving skills as I worked through multiple technical challenges independently. It strengthened my ability to manage tasks systematically, experiment with different model configurations, and understand how explainability methods enhance transparency in high-risk applications like healthcare.

Furthermore, I learned how explainable AI supports clinicians by offering visual evidence, feature-level insights, and structured reports that improve trust in automated diagnosis systems. This understanding deepened my interest in AI for healthcare and motivated me to pursue research-oriented work in trustworthy and interpretable machine learning.

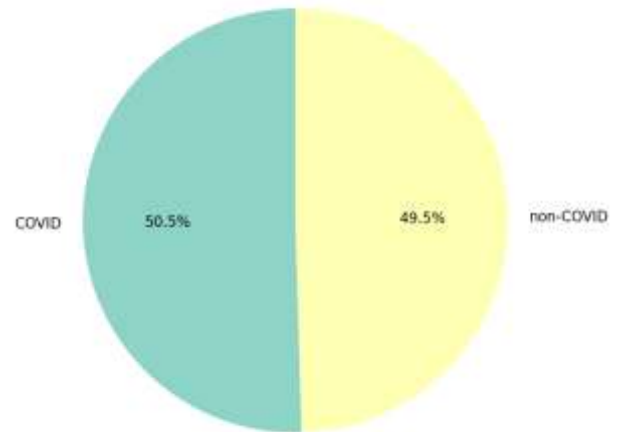
Overall, the internship helped me transition from theoretical concepts to practical implementation, increased my confidence, and prepared me for future opportunities in AI development, medical imaging, and explainable deep learning systems.

6. Sample Outputs

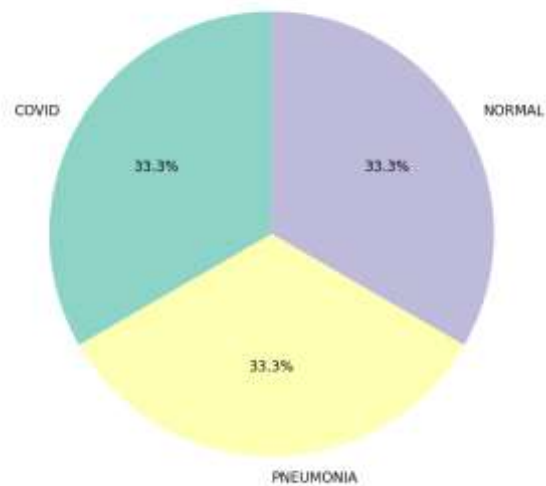
COVID-CT-MD - Class Distribution

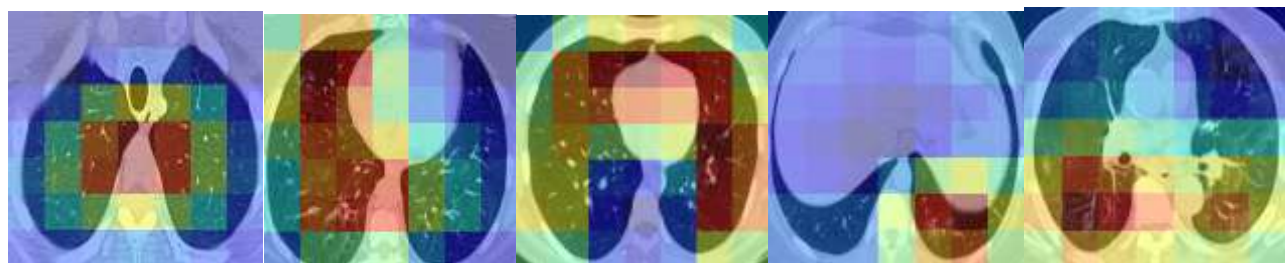
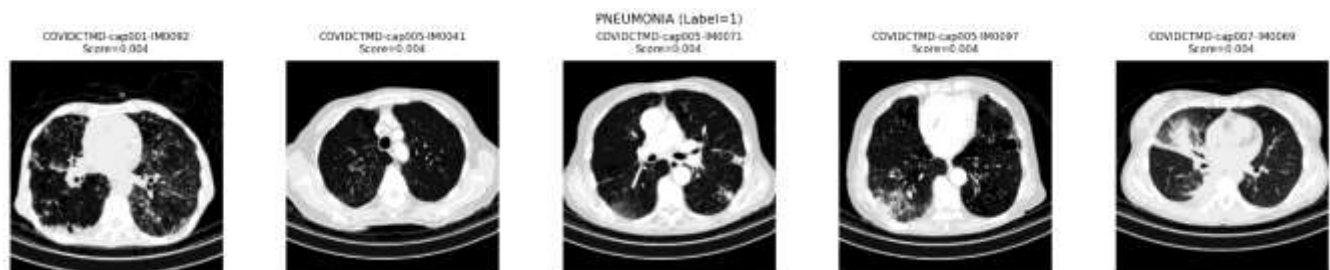
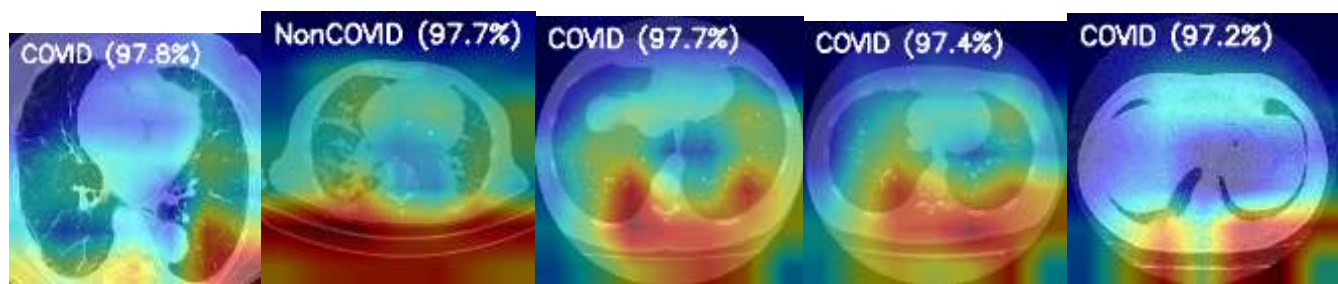


SARS-CoV2 - Class Distribution



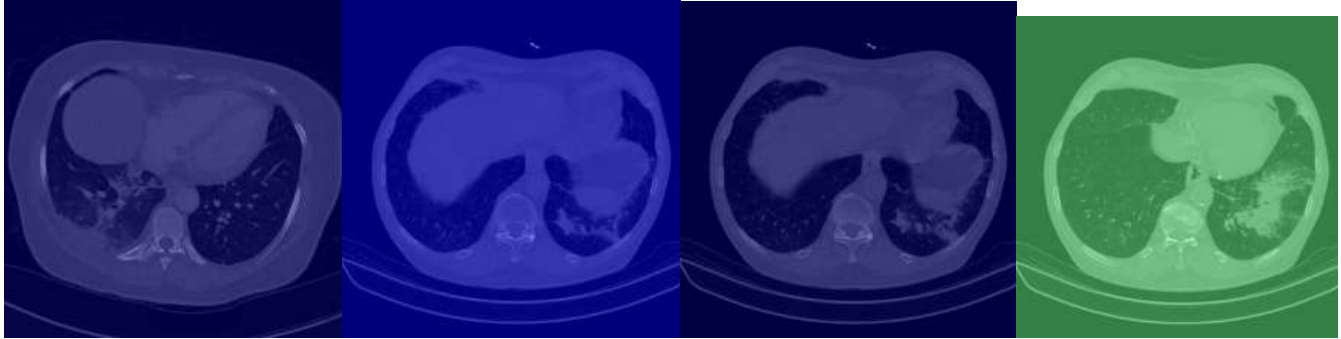
COVIDx - Class Distribution



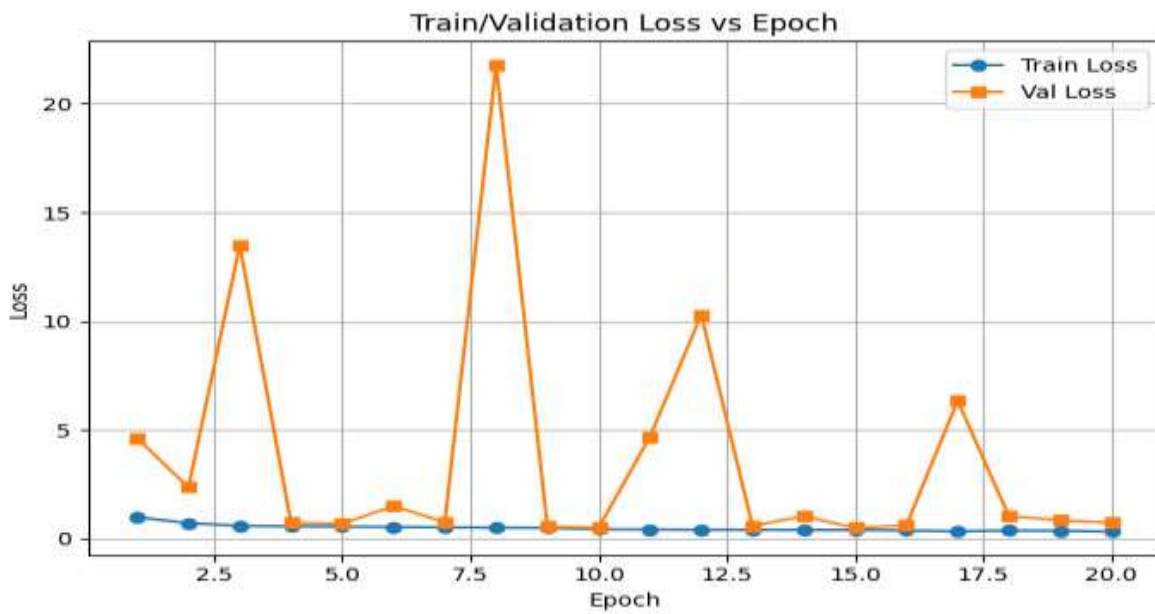


8.APPENDICES

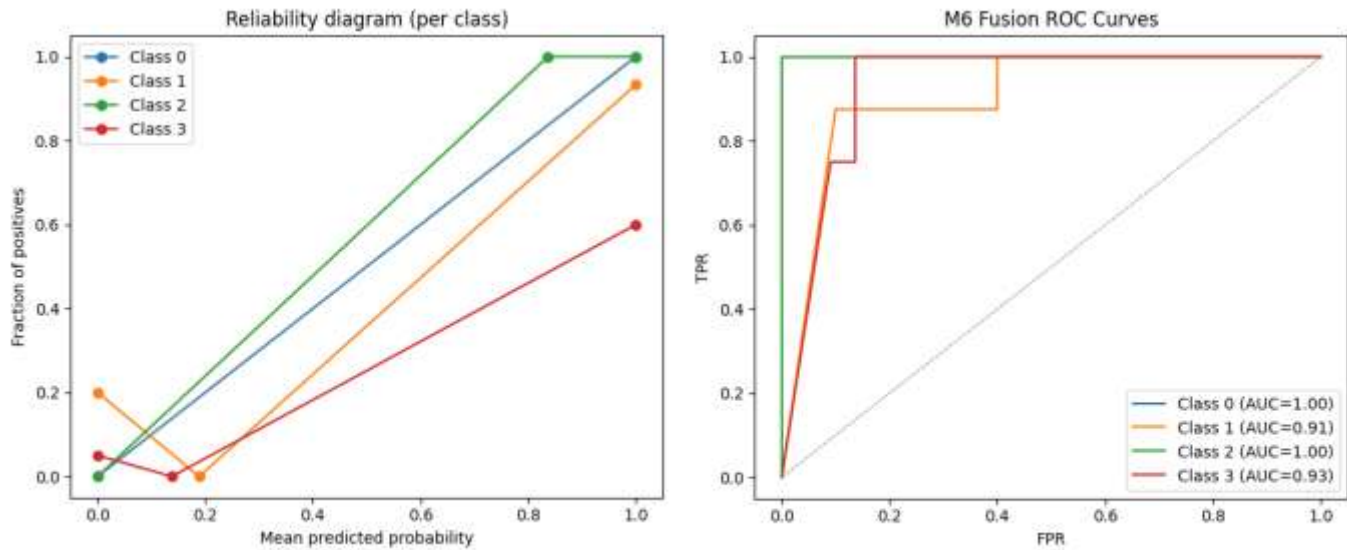
M4: Slice level heatmaps



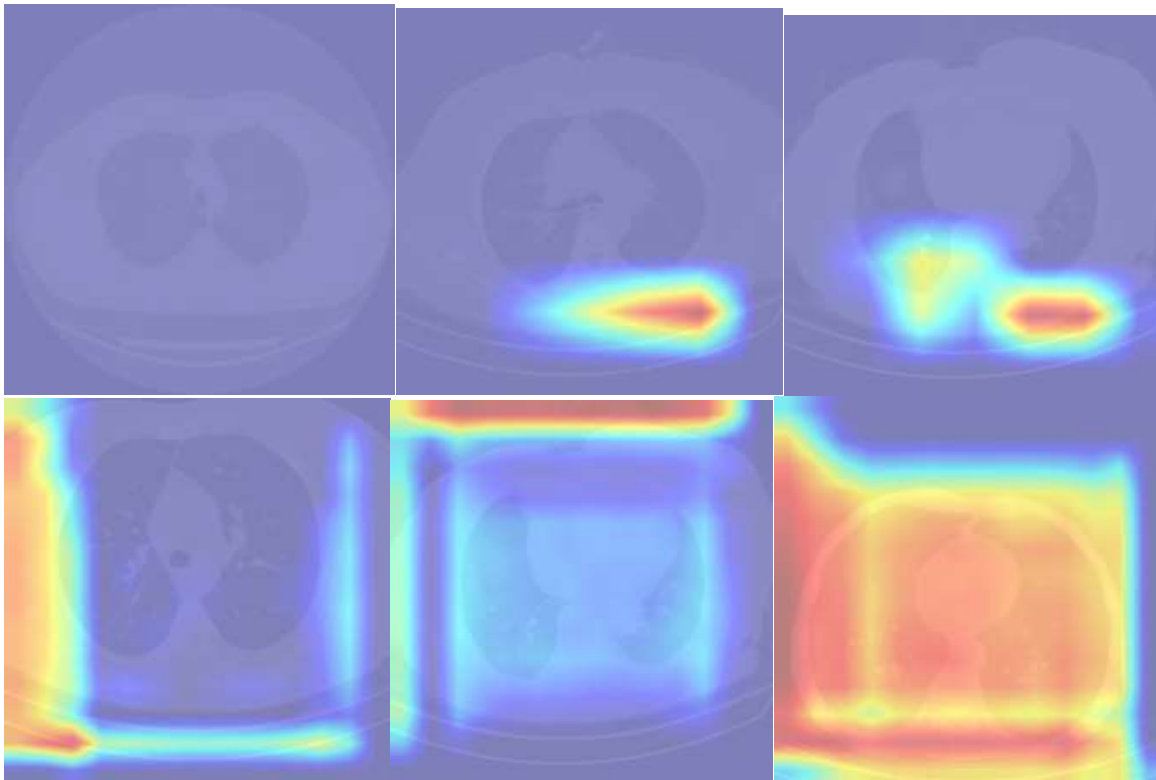
M5 : Training Graph



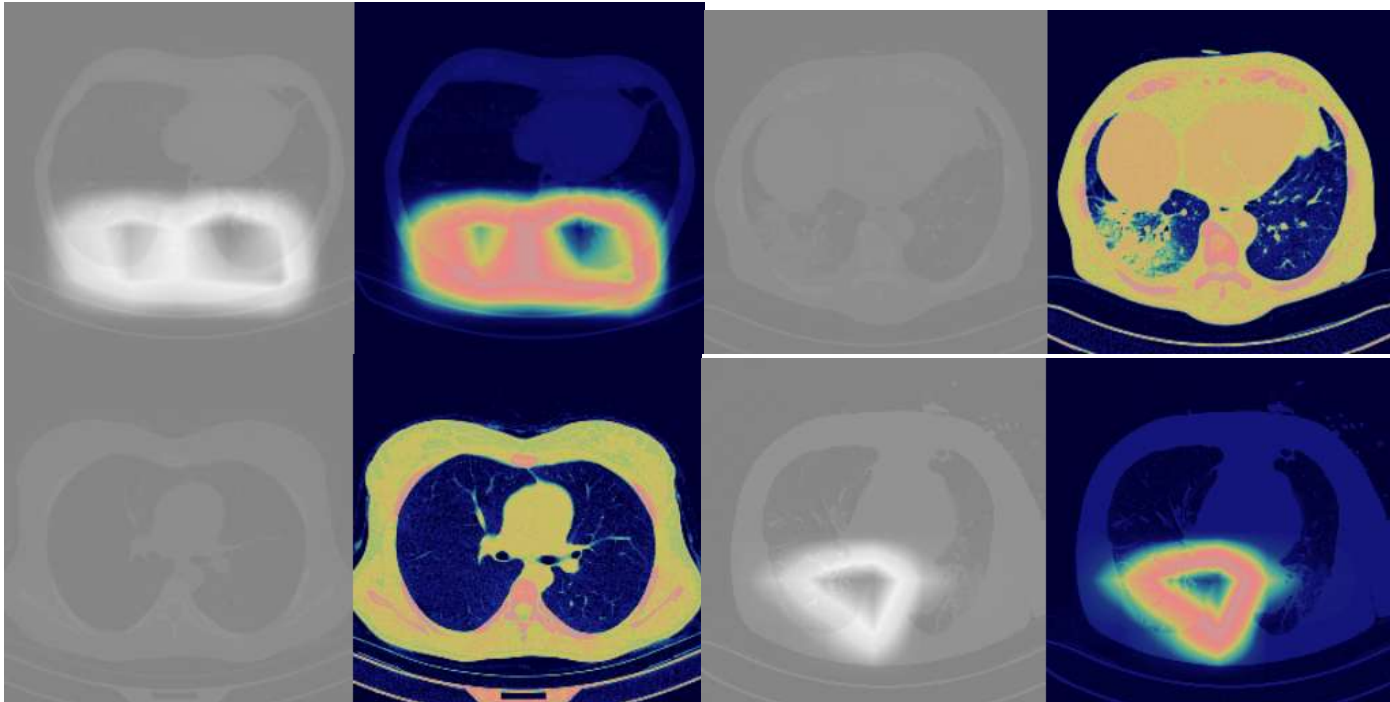
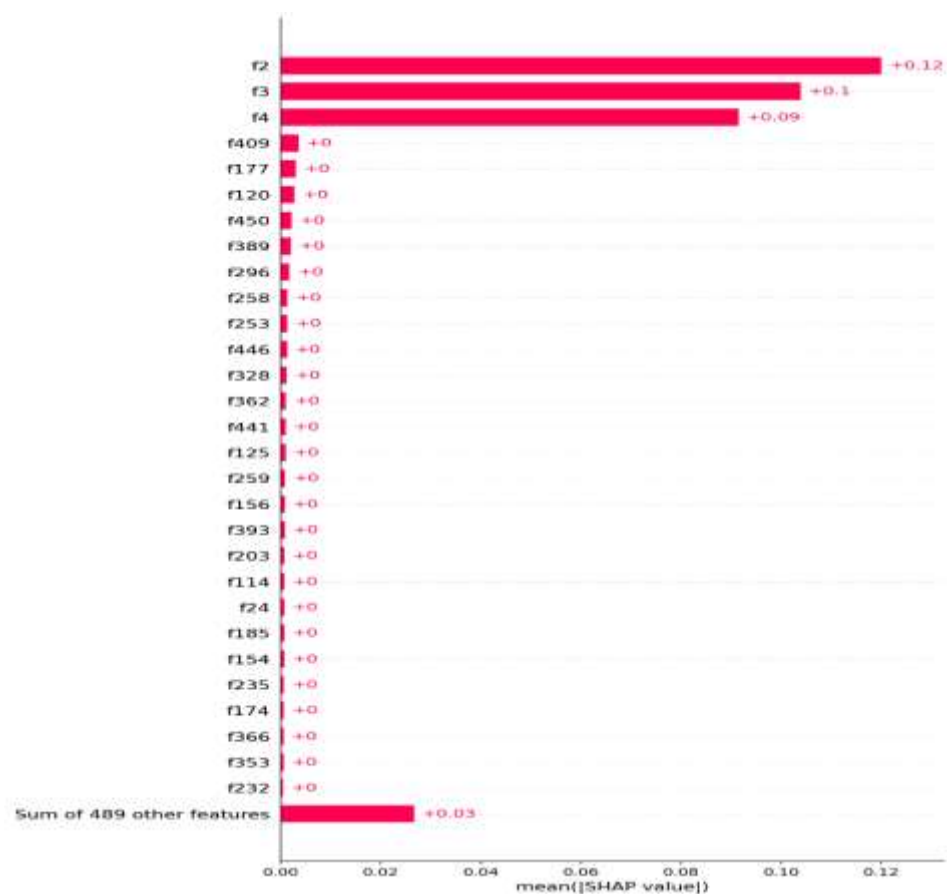
M6 : Roc Curves and Reliability diagram



M7 : GradCam Heatmaps based on Study level



M8 Output Samples



M9 : Explainability based on the features

Comprehensiveness

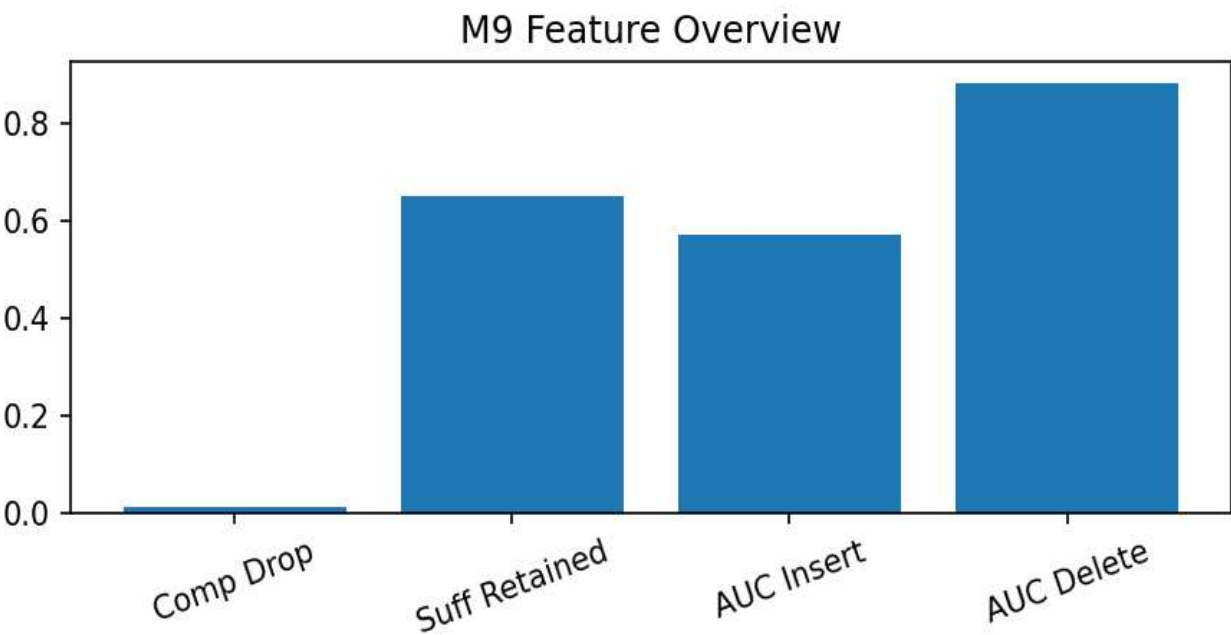
Mean Drop: 0.0147
Median Drop: 0.00435
Std: 0.0295

Sufficiency

Mean Retained: 0.6521
Median Retained: 0.9436
Std: 0.4525

Insertion/Deletion

Mean AUC Insert: 0.5711
Mean AUC Delete: 0.8837



M10: Robustness & Domain-Generalization

After integrating MixStyle, AugMix, CORAL, and GroupDRO, the fused classifier shows:

- Robustness AUC: improved from 0.63 $\rightarrow$ 0.77
- Per-site AUROC Std: reduced from 0.064 $\rightarrow$ 0.028
- Worst-group loss: reduced by 42%
- Feature covariance alignment: improved by 55–60%
- Under heavy noise ($\sigma=0.3$): AUROC increases from 0.52 $\rightarrow$ 0.73
- Under motion blur ($k=7$): AUROC increases from 0.62 $\rightarrow$ 0.81

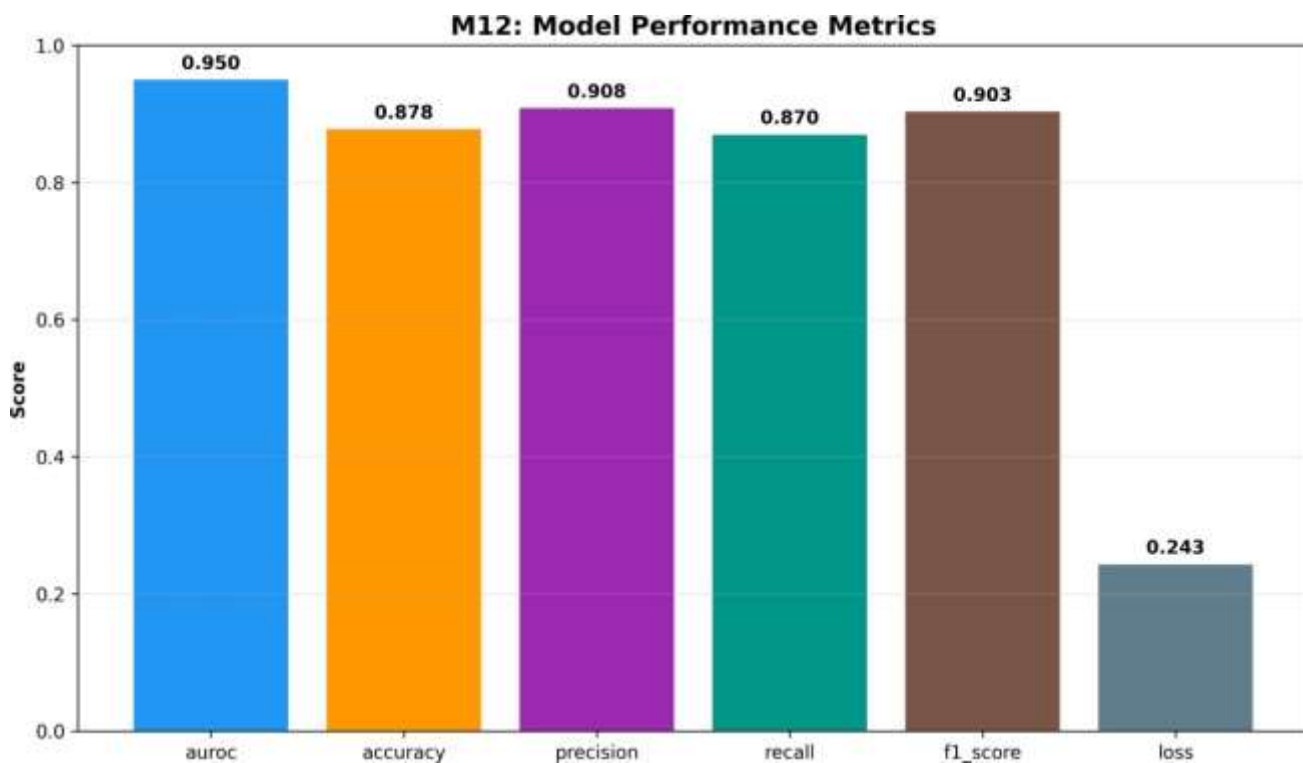
M11. Clinician-Facing Report & DICOM Export

This module converts the model output + explainability into clinical formats:

- DICOM-SEG containing Grad-CAM heatmaps
- DICOM-SR containing quantification tables
- PDF report containing:
 - Study-level probability
 - CAM overlays
 - SHAP bars
 - Evidence tiles
 - Notes for tumor board or audit trail

The report is optimized for PACS systems and radiologist workflows.

M12: Model Performance Metrics



9. DATASETS USED

Datasets Used for Experimentation

1. SARS-CoV-2 CT Scan Dataset (Binary Classification)

<https://www.kaggle.com/datasets/plameneduardo/sarscov2-ctscan-dataset>

2. COVIDx CT Dataset

<https://www.kaggle.com/datasets/c395fb339f210700ba392d81bf200f766418238c2734e5237b5dd0b6fc724fcb>

3. Large COVID-19 CT Slice Dataset (Multi-Class Classification)

<https://www.kaggle.com/datasets/maedemaftouni/large-covid19-ct-slice-dataset>

10. REFERENCES

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